



Effect of noise on the assessment of displacements computed from accelerations recorded at linear and nonlinear civil engineering structures



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ABSTRACT

The purpose of this study is to assess the effect of noise on the estimations of response displacements of single degree-of-freedom systems representative of simple civil-engineering structures subjected to earthquakes. Linear and nonlinear oscillators, with frequencies between 0.1 and 10 Hz and subjected to an earthquake excitation, were considered. Nine signal/noise ratios were studied for two noise input locations: signal contamination at the excitation and signal contamination at the acceleration response. The methodologies considered to estimate displacements were the following: Trifunac and Lee [17], Converse and Brady [9], Chiu [7], and Boore et al. [4]. Results show that noise affects more the estimation of displacements in nonlinear systems than in linear ones. The study also provides values of the signal/noise ratios for which acceptable estimations of displacements are obtained for both linear and nonlinear systems. The most suitable methods to estimate displacements from acceleration records are also recommended.

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1. Introduction

Displacement estimations of civil engineering structures have been useful for several typical procedures in structural engineering: after an earthquake, some parameters and procedures to assess structural damage rely on estimations of lateral displacements attained by structures [15]; displacements are used in identification of dynamic properties [2,19], Palacios-Quinonero et al. [14] present a design strategy to control the vibration of large-scale structures based on both velocity and displacement estimations; Siami et al. [16] use the structural response, along with a passive vibration isolator methodology based on inerters, to improve the vibration performance of an important statue.

Although direct displacement measurements of structures can be recorded with displacement meters such as LVDTs (linear variable differential transformers), lasers, etc., the use of such devices requires static external supports to maintain part of the equipment at a fixed position. The use of external supports to obtain displacements from acceleration records is common practice. Global positioning systems (GPS) can also be used to estimate displacements. However, the sampling frequencies of such devices

are small (1–20 Hz) and their accuracy is only close to 1.0 cm. Recent studies [13,11] have reported some problems with the use of this technology to record reliable and accurate displacements.

To obtain displacements from accelerometers, the acceleration records should be processed by applying double integration. Without such processing (baseline adjustment and/or filtering), computed displacements may be unreal [7,5]. The (digital) processing of raw acceleration records is intended to reduce the effects of both noise and signal distortion that cause both errors in the baseline and, as a consequence, differences in the computed displacements with respect to the true values. In the context of this study, noise is understood as an unwanted signal that interferes with the measurement of the main signal. Therefore, in a sense, the true signal is unknown. Although signal distortion is a different concept [18], herein it is considered as part of the noise because both limit the accuracy of results obtained from the measurement systems.

A previous study by Arias-Lara and De-la-Colina [1] presents a comparative analysis of several methods to estimate displacements from measured acceleration records, with the objective of assessing their capacity to compute displacements of system excited by different types of movements, such as harmonic, vehicular, pedestrian and seismic. Computed displacements were compared with actual measured displacements. Results indicate the

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following: a) the selection of the best method to estimate displacements depends on: displacement amplitudes, waveform type, required computed displacements (either peak displacement or whole displacement record), and magnitude of the final (residual) displacement; b) estimation of accurate displacements is possible, except for cases where the acceleration signal/noise ratio is small; c) noise plays an important role in the estimation of displacements from acceleration records. Thus, further investigation on noise is required.

In the present work, the effect of noise on the estimations of response displacements is evaluated with computer simulations of linear and nonlinear single-degree-of-freedom (SDOF) systems representative of simple civil structures subject to earthquakes (e.g., buildings or bridges). This is a numerical study and its scope does not include experimental work with large-scale structures. Although SDOF systems are simple dynamic models, they can represent with good accuracy the behavior of multi-story buildings with particular characteristics. The use of SDOF oscillators in the present study is not a size limitation of the structures to be analyzed. Structures with multiple degrees of freedom may be condensed to systems with only one SDOF.

Even though noise may come from different sources, only two cases are investigated in this study: 1) noise originated in the excitation signal (in the earthquake record), and 2) noise originated in the response signal (in the computed response acceleration). The former case is when ambient vibration affects the study of seismological excitations and the response is recorded with free-error accelerometers; the latter case may simulate erroneous readings in the structure response, regardless of the input signal. Although noise can be partly generated at both the input and the output, both extreme cases can provide a broad insight into the effect of noise on the estimation of displacements. In both cases, displacements are computed from the acceleration records that correspond to the computed response of the models.

The study is limited to a frequency range of oscillators that is relevant to common civil engineering structures subject to earthquake ground motions. Specifically, the interval of frequencies for the analyzed oscillators varies between 0.1 and 10 Hz. As a reference, this interval approximately covers typical fundamental frequencies of buildings between 1 and 10 stories.

2. Methods to compute displacements from accelerations

The selected methodologies to estimate displacements from measured accelerations are briefly described in this section.

The method by Trifunac and Lee [17] is based on baseline adjustment, carried out with high-pass filtering of acceleration; the cutoff frequency of the filter is selected from the signal-to-noise analysis. To compute displacements, the acceleration record is integrated twice to get the velocity and displacement data. In order to avoid long periods error due to uncertainties in the initial values of velocity and displacements, both the computed velocity and displacement records are high-pass filtered at each stage of integration.

Converse and Brady [9] propose a method to process strong motion signals and estimate velocity and displacement data. For the baseline adjustment of records, this method has two main steps that are applied to the acceleration record: a least square linear fit and a high-pass filtering. Before filtering, the method recommends the addition of zero pads at both the beginning and the end of the acceleration record. In the end, the filtered acceleration record is doubly integrated to estimate velocity and displacement records.

A method specifically developed to estimate seismic displacements was proposed by Chiu [7]. Its main purpose was to fix the

errors of the acceleration-record baseline, including those caused by data manipulation. The algorithm consists of three main steps to adjust the baseline: an acceleration-record least squares fit, an acceleration record high-pass filtering, and a subtraction of the velocity initial value. According to Chiu [7], this method reduces the necessary steps for adjusting the baseline errors, thus reducing the side effects due to data manipulation.

Boore et al. [4] propose a methodology to estimate either the permanent or residual displacements from seismic acceleration records by adjusting the baseline. To achieve this, a quadratic fit of velocity by least squares is performed first. Then, the adjustment function is differentiated and subtracted from the acceleration record. Optionally, a high-pass filter can be applied to the acceleration record.

For all the signal processing algorithms, the Simpson's 1/3 integration rule [6] was used to compute velocities and displacements from accelerations.

3. Structural system, earthquake record, and noise

3.1. 3.1. Structural system

Structures are usually analyzed with multiple-degree-of-freedom [MDOF] models [10]; however, when the global structural behavior is the main concern, single-degree-of-freedom [SDOF] models are preferred. In this study, SDOF models are analyzed. A simple system with mass m , initial lateral stiffness k , and viscous-damping coefficient c is considered (Fig. 3.1). This system can be used to analyze the lateral relative displacement u of its mass when it is subject to a lateral base ground acceleration s . For this study, the displacement u , for a selected earthquake record acceleration, is computed with the Structural Analysis Program (SAP). Thus, the computed displacement and its time derivatives (velocity and acceleration) are assumed as “exact” records, in the sense that all their values are not contaminated with noise. The only possible errors in these quantities are those caused by numerical round-off errors and/or by the intrinsic approximations of the algorithms involved in the solution of the differential equation of motion. However, the use of simple models (Fig. 3.1) maintains these errors negligible; therefore, u is assumed as the “exact”, noise-free displacement corresponding to the response acceleration $a = d^2u/dt^2$. This displacement u will be used to assess the effect of noise, as described in the methodology section.

Linear and nonlinear (bilinear) SDOF models are considered to assess the effect of noise on the estimation of displacements. For each case, five initial natural frequencies are considered (Table 3.1). For the nonlinear models, the post-yield stiffness varied from 10% to 30% of the initial stiffness. For all models, a viscous damping equal to 5% was included.

3.2. Earthquake record

Most of the earthquake records are “corrected” before they are used in engineering studies. The correction procedure is intended

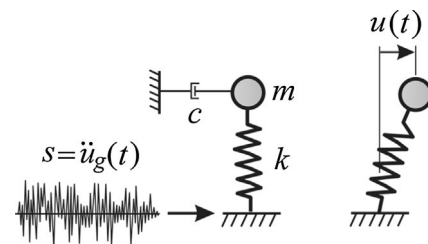


Fig. 3.1. Single degree-of-freedom structural system.

Table 3.1
Single-degree-of-freedom systems.

Model number→	1	2	3	4	5
Initial period T , s	0.2	0.5	1.0	1.5	2.0
Initial frequency f , Hz	5.0	2.0	1.0	0.667	0.5

to remove the noise due to the recording instrument, the baseline adjustment and the background noise [5]. This initial processing of earthquake records can be easily understood with Fig. 3.2, which illustrates a typical uncorrected ground motion record. It can be observed that before the earthquake motion arrives, the instrument is recording background motion.

In this study, an earthquake ground motion is used as the input signal s that excites the base of the structural SDOF systems (Fig. 3.1). Specifically, the input signal s corresponds to the “corrected” north–south (NS) component of the ground motion recorded at El Centro, California during the 1940 Imperial Valley earthquake [8], which is shown in Fig. 3.3. Fig. 3.4 shows its corresponding Fourier amplitude spectrum, which reveals that most of the energy of the signal is in the frequency range of 0.1–10 Hz, approximately.

This corrected record is assumed free of noise. Thus, the response (relative accelerations and displacements) computed with SAP is also assumed noise-free. Noise is inserted in two different sources, both separately and independently, as indicated in the methodology section.

3.3. Noise

As previously mentioned, noise can have different sources, such as instrumental noise, background noise, erroneous initial values of velocity and acceleration, temperature changes, and manipulation. Background noise, which can be seen as ambient vibration, is a major part of the total noise of an acceleration record measured to estimate displacements of civil engineering structures. Ambient vibration can be easily captured with instruments developed for seismological studies. For instance, Fig. 3.5 shows a typical uncorrected acceleration record of ambient vibration on firm soil far from excitation sources (factories, vehicular transit, wind, etc.). Fig. 3.6 illustrates its Fourier amplitude, which shows that the energy of such noise signal is essentially uniform over the frequency range between 0.1 and 100 Hz. Given this almost-uniform distribution of the Fourier amplitude, it is common to use Gaussian white noise (GWN) to model several types of noise [12]. Although there are other sources of noise that can affect the

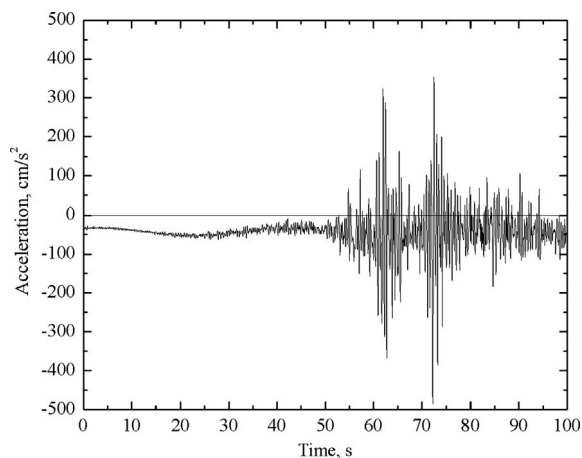


Fig. 3.2. Uncorrected earthquake record.

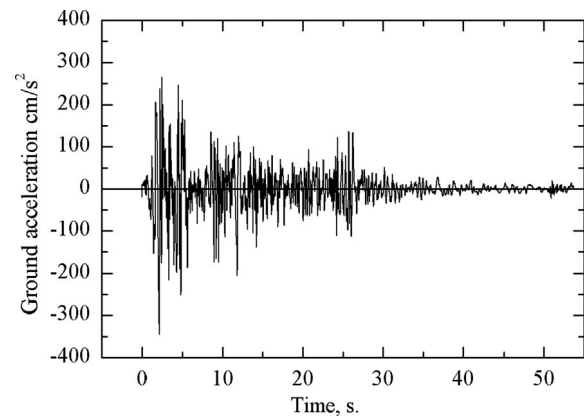


Fig. 3.3. Corrected earthquake ground motion (signal s).

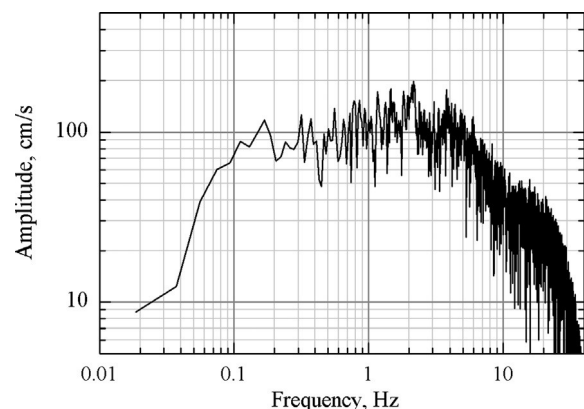


Fig. 3.4. Fourier amplitude of the 1940 El Centro ground motion s .

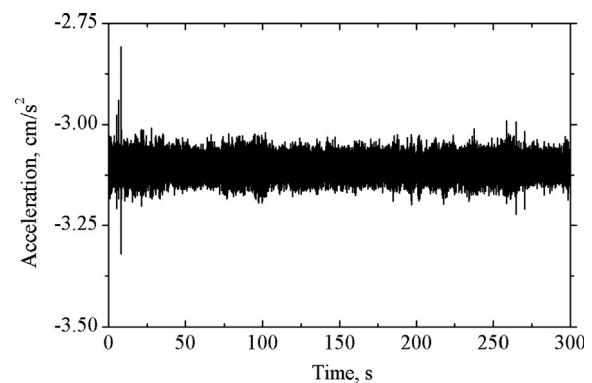


Fig. 3.5. Ambient-vibration, uncorrected acceleration record.

computation of displacements of civil engineering structures, in this study noise is assumed as Gaussian white noise.

4. Methodology

The effect of noise on the computation of displacements is evaluated by computing displacements from acceleration records. Nine signal/noise amplitude ratios were analyzed: $S/N = 10000/1$, $1000/1$, $100/1$, $50/1$, $25/1$, $10/1$, $5/1$, $2/1$ y $1/1$. As for the structural behavior, two cases are studied separately: elastic and inelastic systems. In relation to the source of the noise, two cases are considered: 1) noise generated at the acceleration input signal, and 2) noise generated at the acceleration response. Both cases are

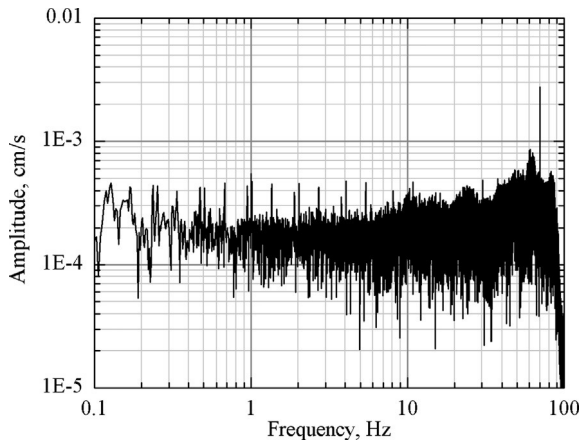


Fig. 3.6. Fourier amplitude of a firm soil, ambient vibration acceleration record.

described below separately (Fig. 4.1). The notation is as follows: s = signal corresponding to the basic input signal (earthquake acceleration record), n = signal corresponding to the Gaussian white

noise, s^* = contaminated input signal (earthquake acceleration record plus noise n), u = oscillator mass displacement computed with the structural analysis program SAP, using s as input, a = oscillator mass acceleration computed with the structural analysis program SAP, using s as input, u^* = oscillator mass displacement computed with the structural analysis program SAP, using s^* as input, a^* = oscillator mass acceleration computed with the structural analysis program SAP, using s^* as input, $\hat{u}$ = estimated oscillator mass displacement computed with methodologies of Section 2, using a^* as input, a' = contaminated response signal (acceleration response a plus noise n), $\tilde{u}$ = estimated displacement computed with methodologies of Section 2, using a' as input.

Case 1 – Noise included in the input signal. To assess the effect of noise at the input signal, it is initially assumed that the signal s is free of noise. In fact, this is usually not true for actual earthquake ground motions because the recorded motion can be contaminated with ambient vibration or other errors caused by processing the signal; however, this fact is irrelevant for the computation of reference accelerations a and displacements u of the structural system. Several levels of Gaussian white noise n are used to modify s , obtaining contaminated signals $s^* = s + n$, which are then used to compute the response signals a^* and u^* with SAP. It is important

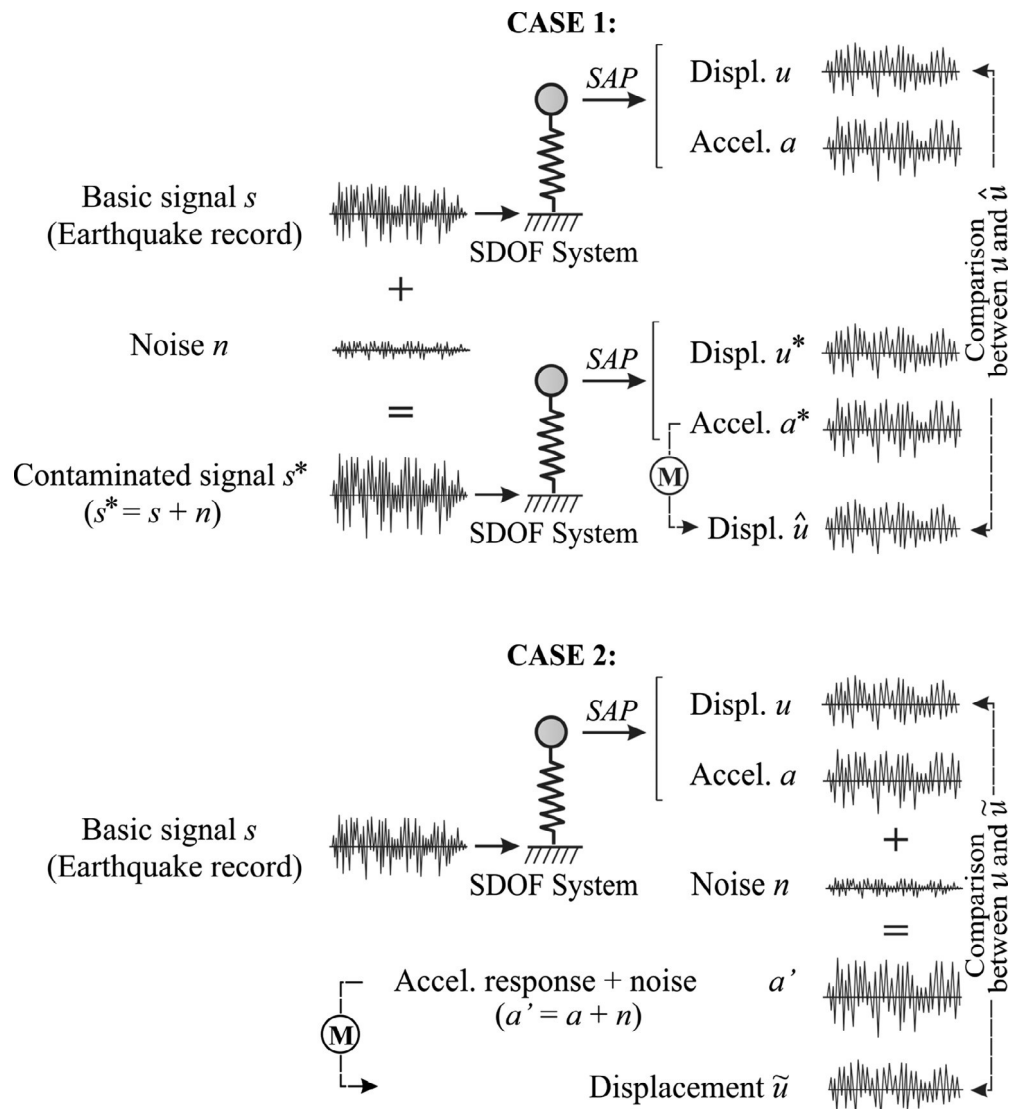


Fig. 4.1. Two cases to estimate the effect of noise. The M inside the circle refers to the methods described in section 2 to estimate displacements.

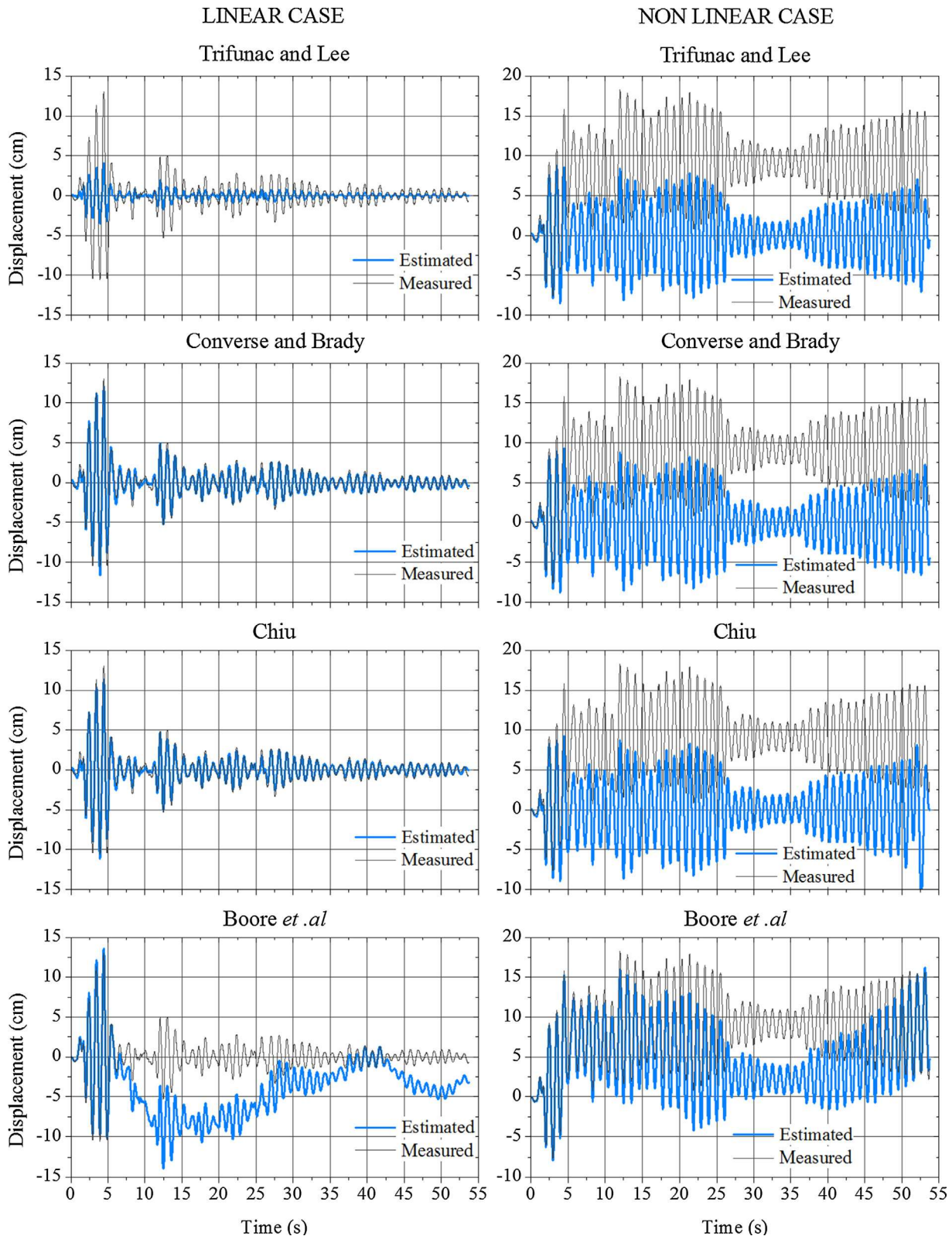


Fig. 5.1. Displacement records for oscillators with $T = 1$ s and signal/ratio = 20 on Case 1.

to highlight that the insertion of noise n in s makes the response signals a and u different from a^* and u^* , respectively. The methods described in section 2 are used to obtain $\hat{u}$ from a^* . It can be observed that the differences between signals u and $\hat{u}$ are due to both the inserted noise n and the algorithms of Section 2. The com-

parison of both signals (u and $\hat{u}$) for different levels of noise reveal the effect of noise on the input signal, for different algorithms proposed to compute displacements from accelerations.

Case 2 – Noise included in the output signal. In this case the signal s is also assumed free of noise. This signal s is used to compute

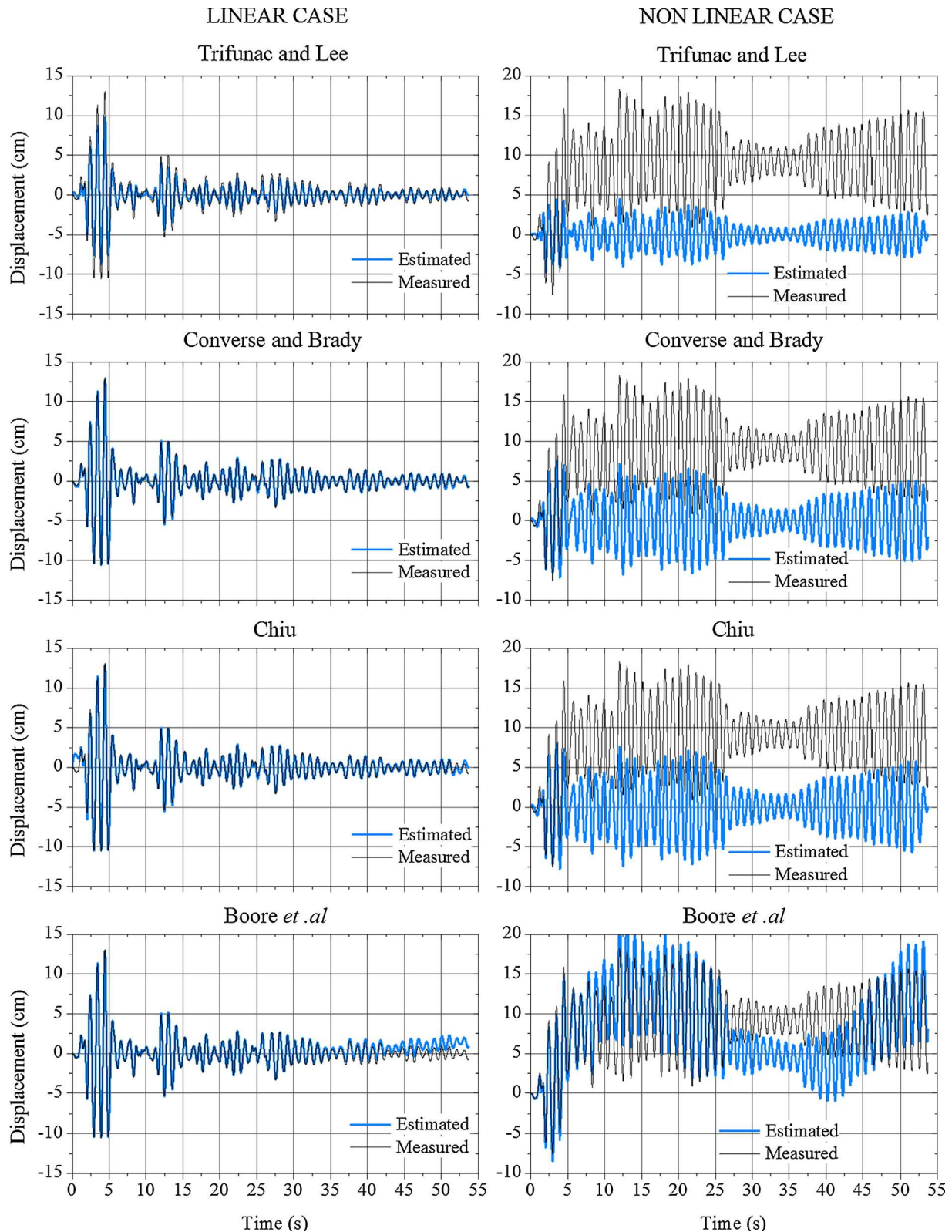


Fig. 5.2. Displacement records for oscillators with $T = 1$ s and signal/ratio = 20 on Case 2.

(with SAP) the response signals a and u . Several levels of Gaussian white noise n are used to modify the computed acceleration a , obtaining contaminated acceleration responses $a' = a + n$. These contaminated signals are used to compute displacements u with the methods described in Section 2. For this case, the observed dif-

ferences between signals u and u are due to both the inserted noise n and the algorithms used to compute displacements (Section 2). The comparison of both signals (u and u) for different levels of noise reveal the effect of noise on the output signal. In general, displacement signals $\hat{u}$ and u are different from each other.

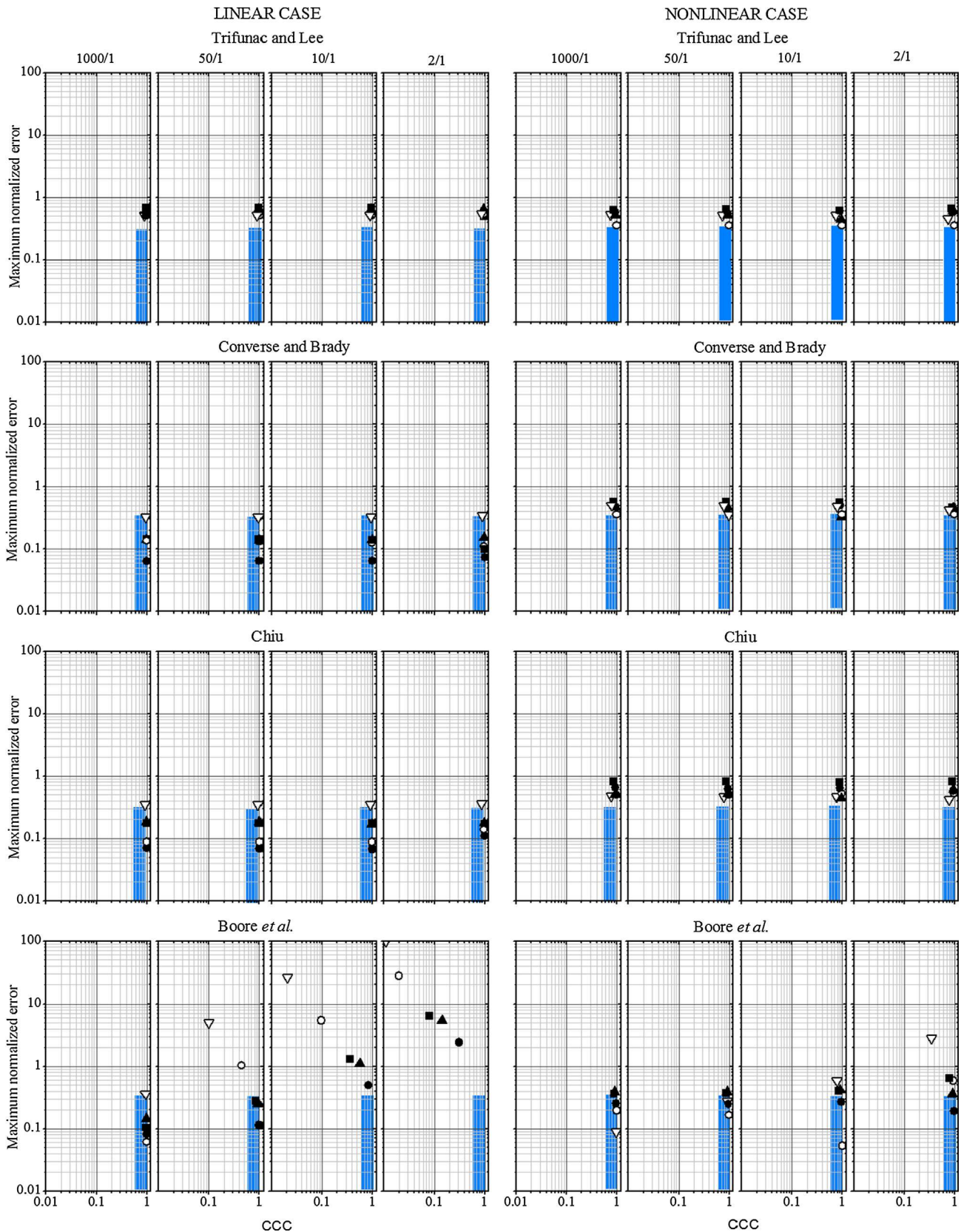


Fig. 5.3. Results (CCC and MNE) for noise as in Case 1, for 4 selected signal/noise ratios of oscillators with initial periods of 0.2 (▽), 0.5 (○), 1.0 (■), 1.5 (▲) and 2.0 (●) seconds.

To assess the effect of noise on displacements, comparison of signals is carried out as follows: for Case 1 (Fig. 4.1), signals u and $\hat{u}$ are compared, while for Case 2, signals u and u are com-

pared. In both cases, displacement u is assumed as the signal of reference. The cross-correlation coefficient (CCC) and the maximum normalized error are used in this study to compare signals. The

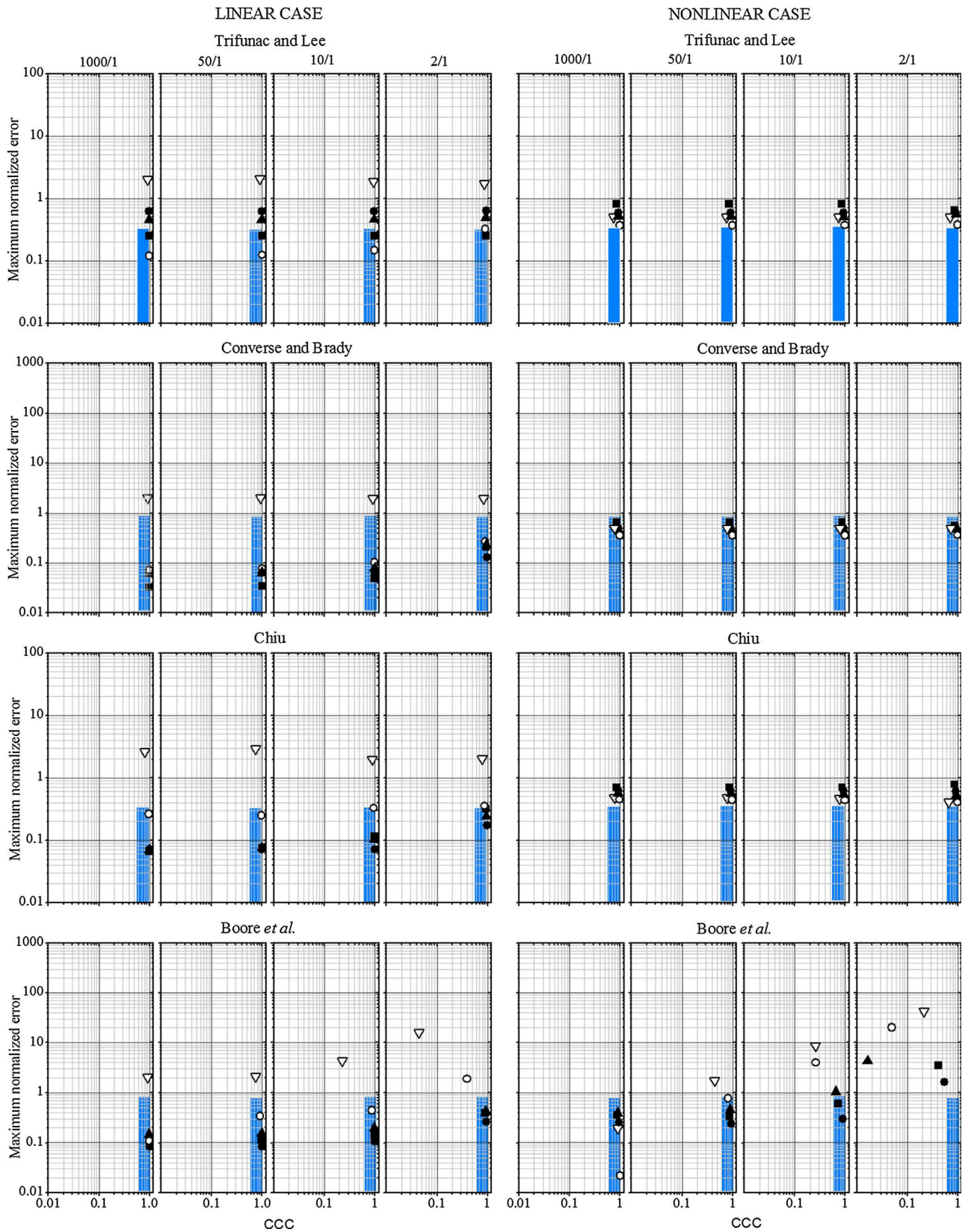


Fig. 5.4. Results (CCC and MNE) for noise as in Case 2, for 4 selected signal/noise ratios of oscillators with initial periods of 0.2 (▽), 0.5 (○), 1.0 (■), 1.5 (▲) and 2.0 (●) seconds.

cross-correlation coefficient [3] is used to determine the similitude between signals x and y ; such coefficient is computed with the following formula:

$$CCC = \frac{1}{N-1} \sum_{i=1}^N \left(\frac{x_i - \bar{x}_m}{\sigma_x} \right) \left(\frac{y_i - \bar{y}_m}{\sigma_y} \right) \quad (4.1)$$

Table 5.1
Identification and efficacy (in percentage) of the studied methods.

Identification and Case $\mapsto$ Method $\downarrow$	Case 1 linear	Case 2 linear	Case 1 nonlinear	Case 2 nonlinear
Trifunac and Lee [17]	0%	38%	0%	0%
Converse and Brady [9]	93%	73%	4%	0%
Chiu [7]	78%	69%	0%	0%
Boore et al. [4]	33%	58%	44%	24%

where: σ_x and σ_y = standard deviations of signals x and y , respectively N = length of signals x_m and y_m = mean values of signals x and y ; $x_i = x(i)$ = i -th element of signal x ; $y_i = y(i)$ = i -th element of signal y

It is well known that CCC varies between -1.0 and 1.0 . A value close to 0.0 indicates no correlation between signals, while a value close to 1.0 indicates good correlation.

The correlation coefficient is a measure of the linear dependence between both signals and indicates the similitude between them. Since a large value of CCC can be obtained even for signals with different amplitudes (e.g., $y(i) = x(i) + c$, being c a constant value), it is also important to keep track of the difference between peaks of the signals to assess the estimation of displacements.

To assess the difference between signal magnitudes, the maximum normalized error (MNE) is used. This error is computed with the following formula:

$$MNE = \frac{\max_t \{x_t - y_t\}}{|\max_t \{x_t\}|} \quad (4.2)$$

where: x_t = value of the reference signal at time t ; y_t = value of the estimated signal at time t

A value close to zero indicates that the normalized difference between amplitudes of both signals is small, for all t . A value of MNE close to ± 1.0 suggests that the largest amplitude difference between signals is almost equal to the maximum amplitude of the reference signal.

5. Results

To simplify the presentation of results in this section, the methods are identified with two capital letters as follows: Trifunac and Lee = TL, Converse and Brady = CB, Chiu = CH, and Boore *et al.* = BO.

As an introduction, Fig. 5.1 shows displacement records for an oscillator with period $T = 1.0$ s and a signal/noise ratio = 20, for Case 1. Each graph compares estimated displacements to exact values. For linear oscillators with final displacements that tend to zero, only the methods of TL, CB and CH provide acceptable estimations of displacements. For nonlinear oscillators, the methods considered do not lead to good estimations of displacements. However, the method of BO provides displacements with the smallest relative differences between signals, particularly for large amplitude displacements.

Fig. 5.2 shows displacements for the same oscillator and signal/noise ratio of Fig. 5.1, but for Case 2. It can be observed that the four methods provide displacement records similar to the reference records. For nonlinear oscillators, the same trend observed for Case 1 is also shown.

Given the large number of oscillator periods and signal/noise ratios considered for Case 1 and Case 2 (linear and nonlinear), showing all the results in terms of the time for the four methods is not convenient. Instead, only condensed results are presented.

5.1. Methods

The efficacy of the tested methods was assessed by computing the percentage of successful displacement estimations for different signal/noise ratios and oscillator periods. For this purpose, it was assumed that a successful estimation simultaneously satisfies

two conditions: $|CCC| \geq 2/3$ and $|MNE| \leq 1/3$, where CCC = cross correlation coefficient and MNE = maximum normalized error. Results were analyzed for each Case as well for linear and nonlinear systems.

Figs. 5.3 and 5.4 show the values of CCC and MNE for oscillators in Case 1 and Case 2, respectively, for selected values of signal/noise ratios. The areas of successful estimations are indicated with the small shaded areas.

For Case 1 with linear oscillators (left side graphs in Fig. 5.3), the best method in terms of the largest percentage of successful displacement estimations was that of CB, with an efficacy of 93%. The second best was the method of CH with 78%, and the third best was the method of BO with 33%. The smallest efficacy resulted for the method proposed by TL with 0%. Although the graphical results of Figs. 5.3 and 5.4 only show some signal/noise ratios, the numerical results reported in this section are computed for all ratios indicated in section 3.

For Case 2 with linear oscillators (left side graphs in Fig. 5.4), the ranking order was the same as for Case 1 (linear behavior). The percentages of efficacy were as follows: CB 73%, CH 69%, BO 58% and TL 38% (Table 5.1).

For Case 1 with nonlinear behavior of oscillators, the BO method was the most successful, with an efficacy percentage of 44%; the CB method had an efficacy percentage of 4%, and the other two methods did not have any successful case.

For Case 2 with nonlinear behavior of oscillators, the only method that registered successful cases was the BO method, with an efficacy percentage of 24%.

It can be observed, that the Converse and Brady method (CB) was the most reliable to estimate displacements of linear oscillators (93% and 73%, for Case 1 and Case 2, respectively), while the BO method was the most reliable to estimate nonlinear displacements (44% and 24%, for Case 1 and Case 2, respectively).

5.2. Linear vs nonlinear oscillators

In a comparison between linear and nonlinear cases, it is observed that the efficacy percentages of linear cases are higher than those of nonlinear cases. This corroborates that it is easier to estimate displacements of linear oscillators than to estimate those of nonlinear oscillators.

5.3. Sources of noise

The efficacy percentages are higher when the noise enters into the systems at the excitation signal (Case 1) than when the noise enters at the response signal (Case 2). This occurs for both types of behavior, linear and nonlinear. Such phenomenon may be due to the participation of the oscillator itself. In Case 1, oscillators work as filters that reduce the noise that is transferred to the response signals used to estimate displacements. In contrast, in Case 2 the oscillators do not participate in the filtering of the noise.

5.4. Signal/noise ratios

To assess the effect of the signal/noise ratio on displacements estimated with the best ranked methods, the efficacy for a

particular signal/noise ratio was calculated for all oscillator periods. Thus, the CB and CH methods were considered, for linear and nonlinear oscillators and for both cases of noise sources (Case 1 and Case 2).

For linear oscillators in Case 1, it was observed for the two best ranked methods (CB and CH) that the only oscillator period for which the displacement estimation efficacy was low corresponded to $T = 0.2$ s. For this oscillator period, the efficacy of CB resulted equal to 67%, while for CH it resulted equal to 0.0%. For all the other oscillator periods, the efficacy of both methods was higher than 90%.

For linear oscillators in Case 2, the CB method showed high efficacy of displacement estimations for $T \geq 0.5$ s, while the CH method showed efficacy for $T \geq 1.0$ s. The efficacy of the CB method for $T = 0.2$ s was zero for all S/N ratios, while for the other oscillator periods, it was close to 90%. For the CH method, the efficacy was equal to zero for $T = 0.2$ s and 66% for $T = 0.5$ s, whereas for the other periods, the efficacy was close to 90%. In both cases (CB and CH), the efficacy of the displacement estimations remained uniform for S/N ratios between 2 and 10000. Most of the estimation failures occurred for S/N ratios = 1.0, for both methods (CB and CH).

For nonlinear oscillators in Case 1, the best ranked method for this case (BO) showed efficacy levels close to 45% and 80% for oscillators with periods $T = 0.2$ s and $T = 0.5$ s respectively; while for $T = 1.0$ s and $T = 1.5$ s, the efficacy was equal to zero. The best performance of this method occurred for $T = 2.0$ s, in which the efficacy was equal to 100%. For $T = 0.2$ s, the method exhibited good performance for S/N ratios ≥ 50 , while for $T = 0.5$ s, the good performance resulted for S/N ratios ≥ 5 .

For nonlinear oscillators in Case 2, the best ranked method (BO) exhibited efficacy levels close to 20%, 20%, 10%, 0% and 70% for periods $T = 0.2$ s, 0.5 s, 1.0 s, 1.5 s and 2.0 s, respectively. The successful estimations occurred for S/N ratios ≥ 1000 of oscillators with $T = 0.2$ s and 0.5 s, as well as for S/N ratios ≥ 10 of oscillators with $T = 2.0$ s.

For linear oscillators, the efficacy of displacement estimations remained approximately constant regardless of the signal/noise ratio. The only case in which the efficacy for two analyzed methods had a considerable decrement was for a signal/noise ratio equal to 1.0. For all the other signal/noise ratios considered in this study the efficacy was almost the same.

For nonlinear oscillators, the BO method was considered because it was the best ranked. For Case 1, three efficacy levels were identified. The first level, with the highest efficacy was identified for signal/noise ratios between 10,000 and 50; the second level was identified for signal/noise ratios between 25 and 5; and the third level, with the lowest efficacy, corresponded to signal/noise ratios smaller than or equal to 2. The efficacy obtained for the first level was approximately three times the efficacy of the third level.

5.5. Sensitivity of results to the system parameters

To analyze the sensitivity of the methods to estimate displacements to the system parameters, the influence of two system parameters (viscous damping ratio and post-yield stiffness) was analyzed. Both parameters were varied as follows: a) damping ratio: 2%, 5% and 10%; and b) post-yield stiffness: 5%, 10% and 20% of the initial stiffness. Linear and nonlinear (bilinear) SDOF oscillators with fundamental periods $T = 0.5$, 1.0 and 1.5 s were considered. In all cases only one parameter was varied at each time.

The analysis shows that, for linear and nonlinear oscillators, both coefficients (viscous damping and post-yield stiffness) do not have a significantly influence on the computed displacements.

Small variations of both the cross-correlation coefficients (CCC) and the maximum normalized errors (MNE) are obtained when the structure parameters are varied in the ranges indicated before.

6. Conclusions

In this study, the effect of noise in the computation of displacements from acceleration records was analyzed; several methods to compute displacements were tested as well. The efficacy of the methods was assessed by computing the percentage of successful displacement estimations. For this purpose, it was assumed that a successful estimation simultaneously satisfies two conditions: $|CCC| \geq 2/3$ and $|MNE| \leq 1/3$, where CCC = cross correlation coefficient and MNE = maximum normalized error. This analysis allows to point out the following conclusions:

1. The effect of the level of noise on the efficacy of the methods to estimate displacements in the nonlinear behavior is greater than in the linear case.
2. For the best ranked methods for linear behavior and, for Cases 1 and 2, the ratio S/N for which the efficacy of the methods tends to decrease was approximately S/N = 5. For the inelastic cases, where only the BO method reported successful cases, the S/N ratio for which a significant reduction of efficacy was observed is approximately S/N = 25 for Case 1 and S/N = 100 for Case 2. This means that for similar signal/noise ratios, better estimations of displacements are obtained when the noise enters the excitation signal (Case 1) than when the noise enters the response signal (Case 2).
3. For linear oscillators (Case 1 or Case 2), systems with small oscillation periods ($T < 0.5$ s) are more affected by noise than systems with large oscillation periods. For nonlinear systems, no clear trend of the effect of noise on estimated displacements was observed in terms of the oscillation period.
4. The Converse and Brady [9] [CB] method was the most successful method in the estimation of displacements of linear oscillators, while the method of Boore et al. [4] [BO] was the most successful method for nonlinear oscillators.

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